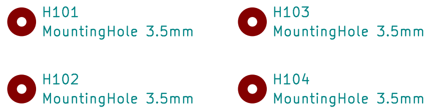
MQTT - Modular Controller Board

Summary:

ESP32 S3 based controller for attached modules. Designed to receive commands via MQTT and send required information. This version supports four individual modules being attached. USB C connection for programming, and only supports the controller itself.

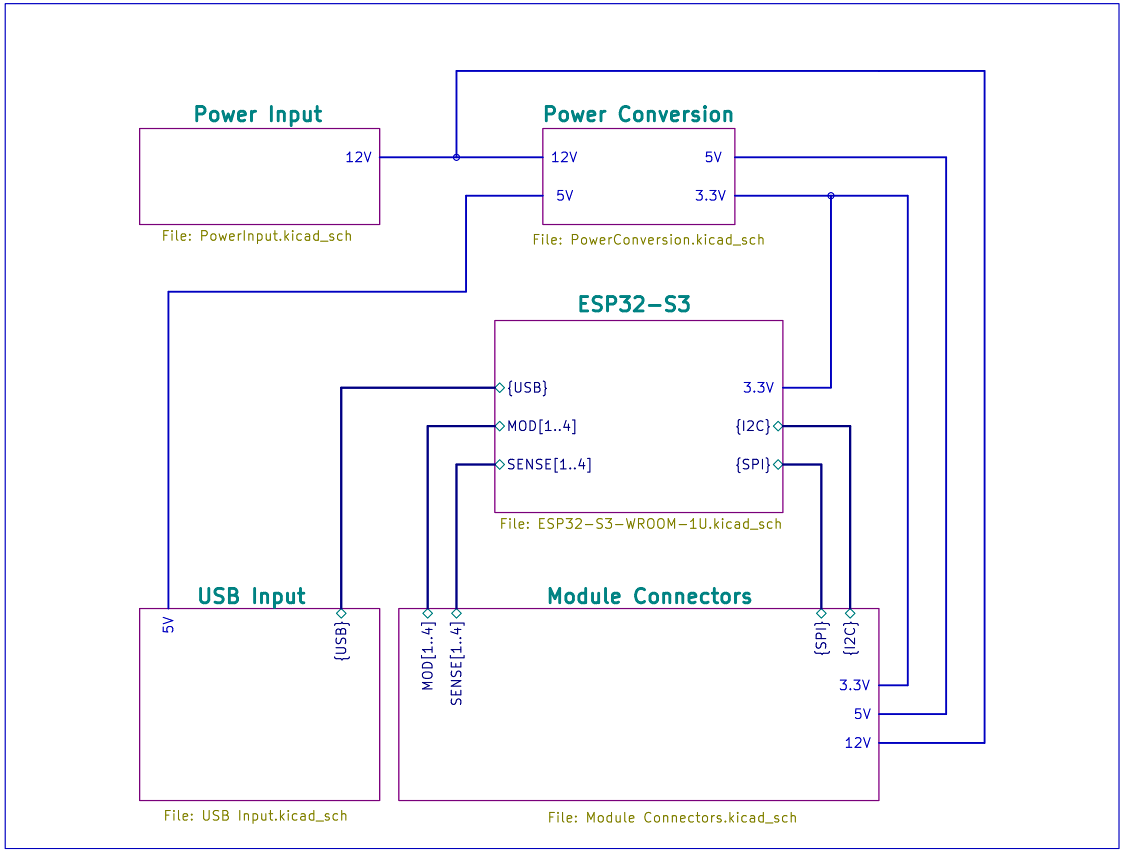
Table of Contents

Non BOM Items



Additional BOM Items

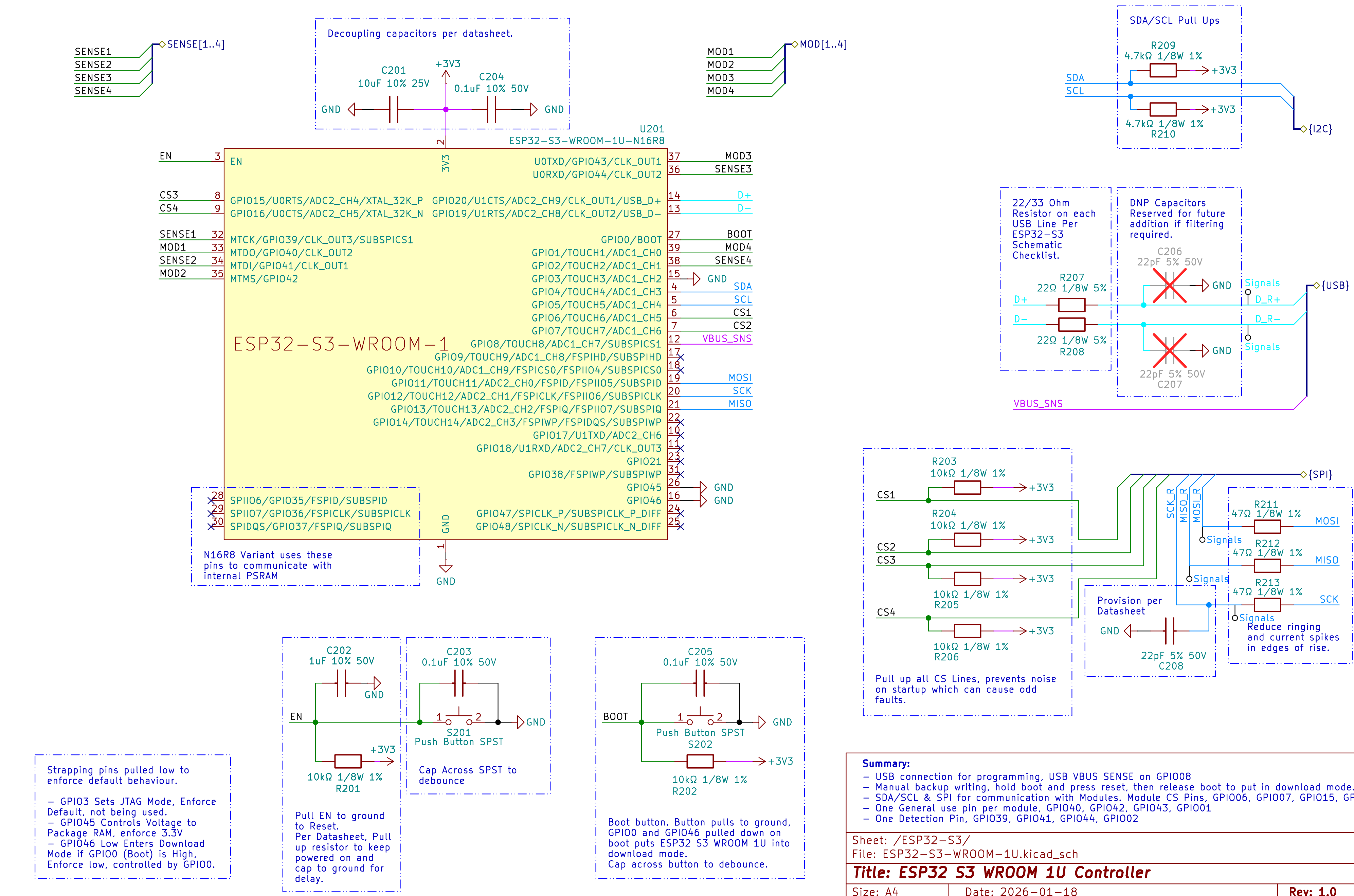
-



Sheet: /
File: MQTT-ModularControllerBoard.kicad_sch

Title: MQTT - Modular Controller Board

Size: A4	Date: 2026-03-14	**Rev: 1.0**
KiCad E.D.A. 9.0.6		Id: 1/6

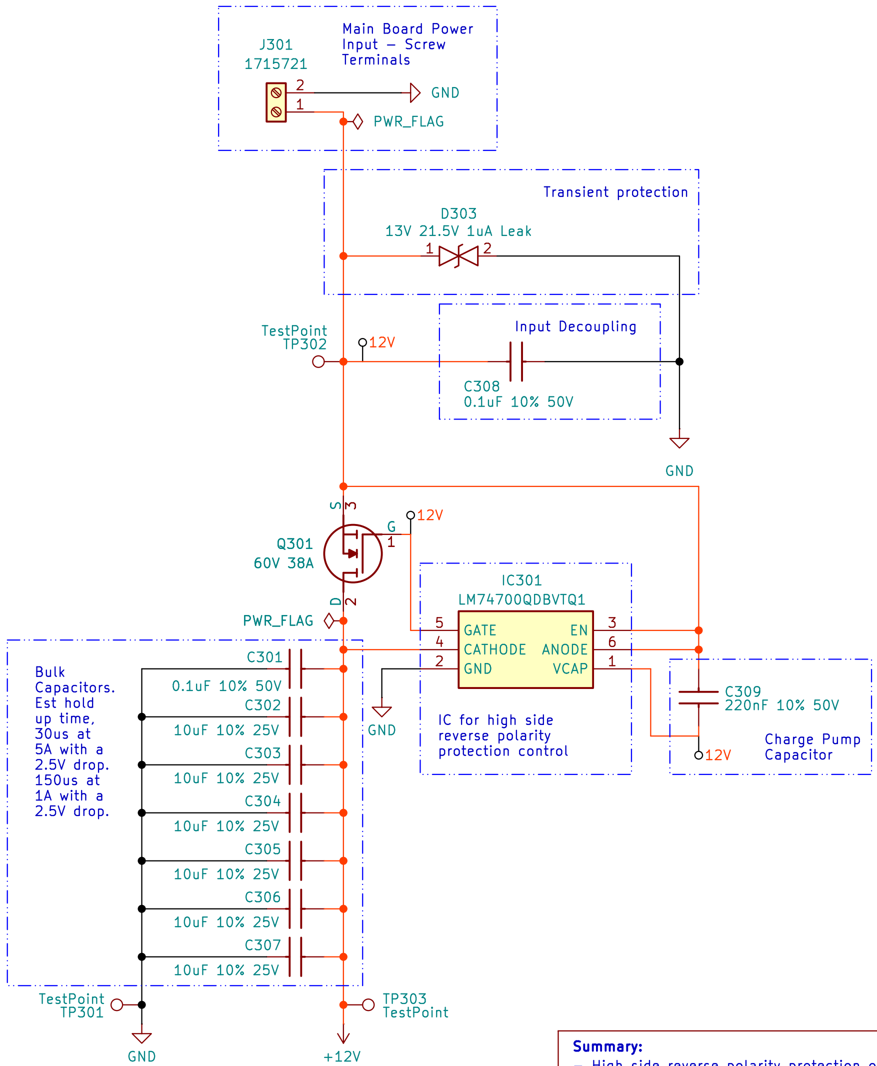


Summary:
- USB connection for programming, USB VBUS SENSE on GPIO08
- Manual backup writing, hold boot and press reset, then release boot to put in download mode.
- SDA/SCL & SPI for communication with Modules. Module CS Pins, GPIO06, GPIO07, GPIO15, GPIO16
- One General use pin per module, GPIO40, GPIO42, GPIO43, GPIO01
- One Detection Pin, GPIO39, GPIO41, GPIO44, GPIO02

Sheet: /ESP32-S3/
File: ESP32-S3-WROOM-1U.kicad_sch

Title: ESP32 S3 WROOM 1U Controller

Size: A4	Date: 2026-01-18	**Rev: 1.0**
KiCad E.D.A. 9.0.6		Id: 2/6

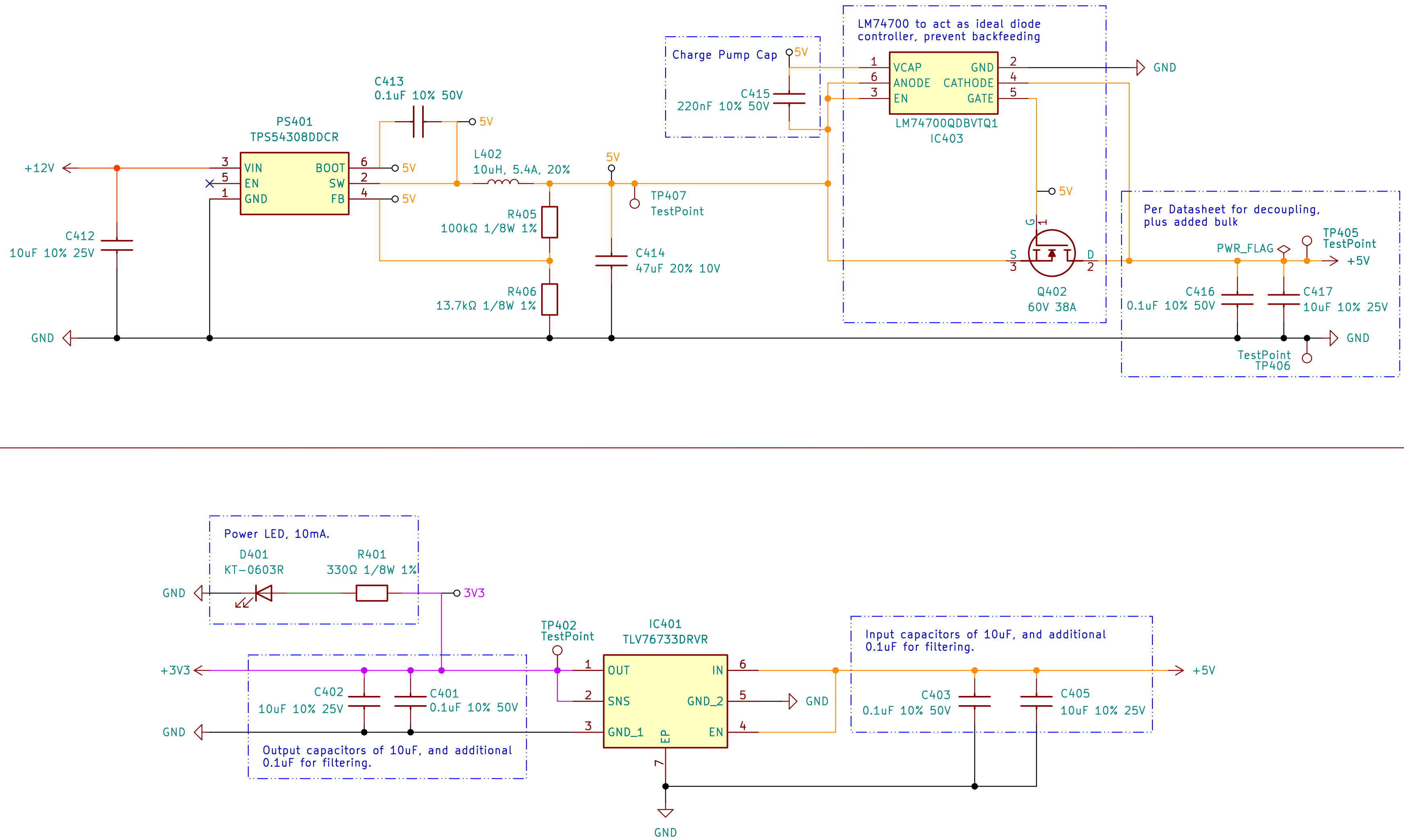


Summary:
- High side reverse polarity protection on 12V input. Safer and only 20mV voltage drop.

Sheet: /Power Input/
File: PowerInput.kicad_sch

Title: Power Input & Protection

Size: A4	Date: 2026-03-15	**Rev: 1.0**
KiCad E.D.A. 9.0.6		Id: 3/6



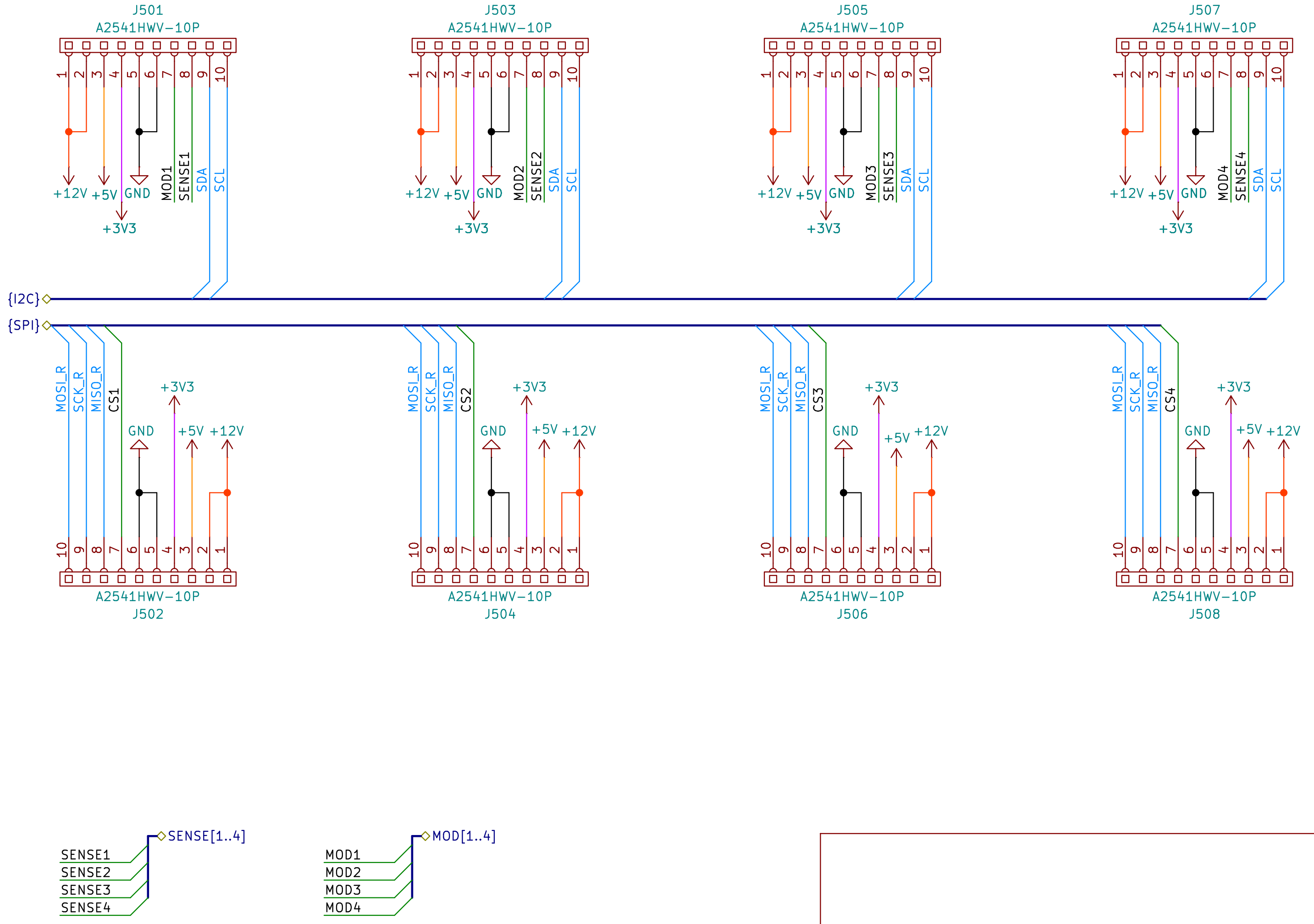
Summary:

- Buck converter generated with TI WEBENCH POWER DESIGNER for 5V output and max 3A output.
- LDO Conversion of 5V Power to 3.3V with a max Peak of 1A. Ideally < 800mA.

Sheet: /Power Conversion/
File: PowerConversion.kicad_sch

Title: Power Conversion

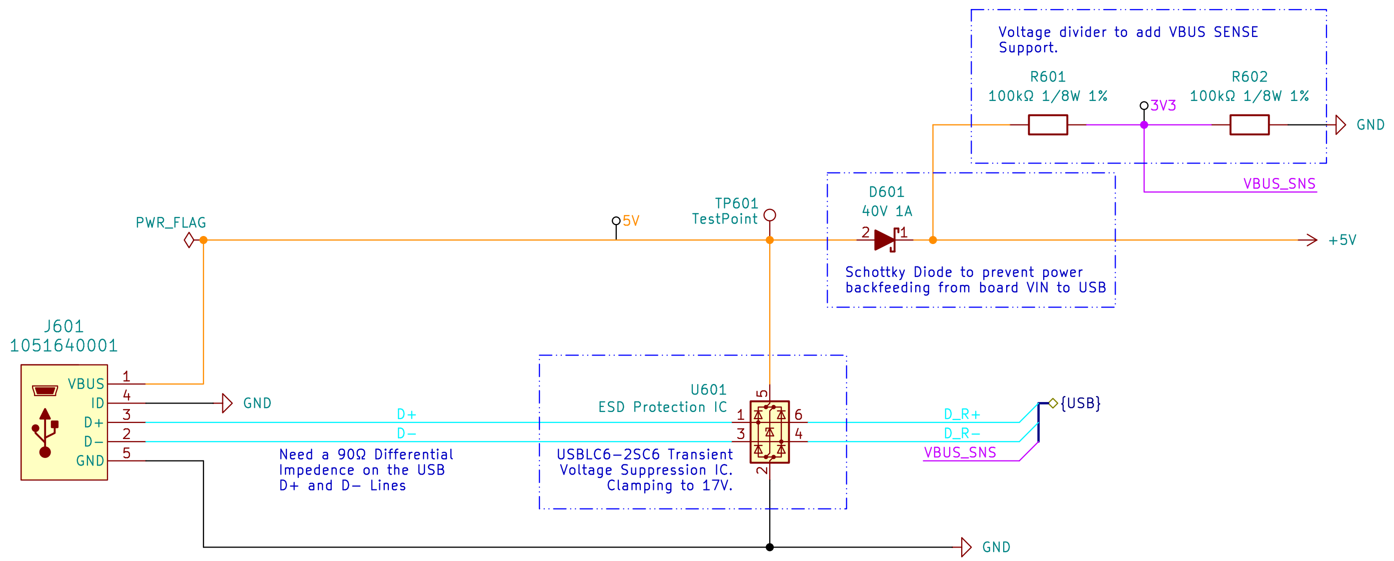
Size: A4	Date: 2026-07-24	**Rev: 1.0**
KiCad E.D.A. 9.0.6		Id: 4/6



Sheet: /Module Connectors/
File: Module Connectors.kicad_sch

Title: Module Connectors

Size: A4	Date: 2026-03-15	**Rev: 1.0**
KiCad E.D.A. 9.0.6		Id: 5/6



Summary:
- VBUS_SNS in place for USB connection while connected to 12V. Requires detection of being self powered on startup in software.

Sheet: /USB Input/
File: USB Input.kicad_sch

Title: USB C Connection

Size: A4	Date: 2026-03-18	**Rev: 1.0**
KiCad E.D.A. 9.0.6		Id: 6/6